

HDOC Day-26

Write a C program that displays the following menu and uses a switch statement to print the “Right-aligned patterns” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameterless function with no return value.

Note: For each activity, prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it against the prepared test cases.

1. Right-Aligned Increasing Triangle

Pattern for n = 4

Algorithm:

1. Start.
2. Input the number of rows n.
3. For i = 1 to n
Print n-1 spaces
Print i stars
4. Move to the next line after every row.
5. Stop.

2. Right- Aligned Increasing Number Triangle

Pattern for n = 4

Algorithm:

1. Start.
2. Input the number of rows n.
3. For i = 1 to n:
Print n-i spaces.
Print i stars.
4. Move to the next line after every row.
5. Stop.

3. Right-Aligned Decreasing Triangle

Pattern for n = 4

Algorithm:

1. Start
2. Input the number of rows n.
3. For i = 1 to n:

Print i-1 groups of spaces.

Print n-i+1 stars.

4. Move to the next line.
5. Stop.

C Program:

```
• #include <stdio.h>
•
• // Function 1: Right-aligned increasing triangle
• void increasingTriangle(int n)
• {
•     int i, j;
•
•     for (i = 1; i <= n; i++)
•     {
•         // Print spaces
•         for (j = 1; j <= n - i; j++)
•         {
•             printf(" ");
•         }
•
•         // Print stars
•         for (j = 1; j <= i; j++)
•         {
•             printf("* ");
•         }
•
•         printf("\n");
•     }
• }
•
• // Function 2: Right-aligned increasing number triangle
• void increasingNumberTriangle(int n)
• {
•     int i, j;
```

-
- for (i = 1; i <= n; i++)
- {
- // Print spaces
- for (j = 1; j <= n - i; j++)
- {
- printf(" ");
- }
-
- // Print numbers
- for (j = 1; j <= i; j++)
- {
- printf("%d ", j);
- }
-
- printf("\n");
- }
- }
-
- // Function 3: Right-aligned decreasing triangle
- void decreasingTriangle(int n)
- {
- int i, j;
-
- for (i = 1; i <= n; i++)
- {
- // Print spaces
- for (j = 1; j < i; j++)
- {
- printf(" ");
- }
-
- // Print stars
- for (j = i; j <= n; j++)
- {
- printf("* ");
- }
-
- printf("\n");

- }
- }
-
- int main()
- {
- int choice, n;
-
- printf("==== RIGHT-ALIGNED PATTERNS =====\n");
- printf("1. Right-aligned increasing triangle\n");
- printf("2. Right-aligned increasing number triangle\n");
- printf("3. Right-aligned decreasing triangle\n");
-
- printf("Enter your choice: ");
- scanf("%d", &choice);
-
- printf("Enter number of rows: ");
- scanf("%d", &n);
-
- switch (choice)
- {
- case 1:
- increasingTriangle(n);
- break;
-
- case 2:
- increasingNumberTriangle(n);
- break;
-
- case 3:
- decreasingTriangle(n);
- break;
-
- default:
- printf("Invalid choice!\n");
- }
-
- return 0;
- }

Test Case Table:

Test Case	Choice	n	Purpose	Expected Result
1	1	4	Valid	Right-aligned rows of 1,2,3,4 stars
2	2	4	Valid	Right-aligned rows: 1; 1 2; 1 2 3; 1 2 3 4
3	3	4	Valid	Right-aligned rows of 4, 3, 2, 1 stars
4	1	1	Boundary	One star
5	2	0 then 3	Invalid then valid	Error, re-input, then 3-row pattern
6	9	-	Invalid choice	Error and menu repeats
7	4	-	Exit	Program terminates

Compilation and Execution:

Test Case 1:

```

(kali㉿kali)-[~/Desktop]
$ ./hdoc26

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 1
Enter number of rows: 4
  *
 * *
* * *
* * * *

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 2
Enter number of rows: 4
  1
 1 2
1 2 3
1 2 3 4

```

Test Case 2:

```

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 3
Enter number of rows: 4
* * * *
 * * *
  * *
   *

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 1
Enter number of rows: 1
*

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 2
Enter number of rows: 2
  1
 1 2

```

Test Case 3:

```
— RIGHT-ALIGNED PATTERNS MENU —  
1. Right-aligned increasing triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing triangle  
4. Exit  
Enter your choice: 9  
Invalid choice. Please try again.
```

```
— RIGHT-ALIGNED PATTERNS MENU —  
1. Right-aligned increasing triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing triangle  
4. Exit  
Enter your choice: 4  
Exiting program.
```